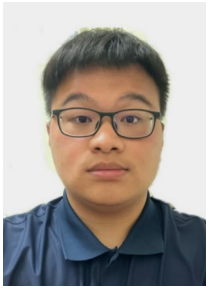
SHUN-JIE FAN



National Cheng Kung University, B.Eng. Engineering Science
Delta Electronics, Software Engineer Intern
Tainan, Taiwan | 0908-238-391 | fanj631@gmail.com
github.com/johnson931128 | johnson931128.github.io

EDUCATION

National Cheng Kung University **Sep 2023 - Jun 2027**

Bachelor of Engineering Science, Department of Engineering Science

- Relevant coursework: Operating Systems (95), Control Systems (94), Algorithms (90), Computer Organization & Assembly (88), VLSI (82).
- Semester average improved from 72.28 in freshman fall to 90.37 in junior spring, with particular interest in system integration, software engineering, and computer architecture.

WORK EXPERIENCE

Delta Electronics - Software Engineer Intern **2026 - Present**

EtherCAT Packet Analyzer | C# | .NET Framework | TShark

- Developing an EtherCAT packet analysis DLL that transforms captured network traffic into structured information for slave-state visualization, protocol analysis, error-flow reconstruction, waveform inspection, and report generation.
- Implemented a pipeline that parses PCAPNG captures, extracts and normalizes EtherCAT datagrams, reconstructs slave states and error-handling sequences, and organizes the results into visual timelines and analysis reports. (In progress)

PROJECT EXPERIENCE

ROS 2 Cross-floor AMR System | **GitHub** **2026 - Present**

ROS 2 Jazzy | Nav2 | SLAM Toolbox | Raspberry Pi 5 | ESP32

- Designed multi-floor navigation and mission control, and built a hardware abstraction layer linking ROS 2 software with motor, encoder, PID, and sensor operations.

C++ 2D AMR Physics Simulator | **GitHub** **2026 - Present**

C++17 | SFML | Differential-drive Simulation | OOP

- Built a lightweight simulator with differential-drive motion, obstacle-map editing, rollback collision handling, external configuration, and modular AMR / Environment / Simulator classes.

Verilog VGA Ping-Pong Game on FPGA | **GitHub** **2025**

Verilog | FPGA | VGA | Digital Logic

- Team Project - Implemented VGA timing, pixel rendering, collision and scoring logic, keypad input, seven-segment display, and dot-matrix output while debugging timing and module synchronization.

Java Maze Game | **GitHub** **2025**

Java | Swing / AWT | OOP | DFS | File I/O | TCP Socket

- Team Project - Developed a GUI maze game with DFS-based random maze generation, save/load state persistence, modular OOP design, and a basic TCP socket multiplayer prototype.

Diabetic Retinopathy AI Screening System **2026**

PyTorch | EfficientNet-B0 | Grad-CAM | Gradio

- Team Project - Trained a referable diabetic-retinopathy screening model with EfficientNet-B0, reaching 0.9183 validation AUC, with class-imbalance handling, Grad-CAM visualization, and a Gradio demo.

SKILLS

Programming: C, C++, C#, Python, Java, Verilog

Systems: Linux, ROS 2, Nav2, SLAM Toolbox

Industrial Communication: EtherCAT

Agent Frameworks: LangGraph

Embedded & Hardware: ESP32, Raspberry Pi 5, 8051, FPGA

Development Tools: Git, CMake, Makefile, colcon, TShark, Vim